The preceding chapters optimized expected returns. This chapter relaxes that assumption in three directions: tracking full return distributions rather than means (distributional RL), imposing constraints on secondary objectives (constrained MDPs), and hedging against model misspecification (robust MDPs).

0.1 Distributional Reinforcement Learning and Risk Measures

Standard reinforcement learning characterizes a policy π by the expected return $V^\pi(s) = \mathbb{E}^\pi\Big[\sum_{t=0}^\infty \gamma^t R_t \mid S_0 = s\Big]$. Distributional reinforcement learning instead maintains the full random variable $Z^\pi(s,a) = \sum_{t=0}^\infty \gamma^t R_t$ whose expectation is $Q^\pi(s,a)$. Bellemare et al. (2017) showed that the Bellman equation has a distributional counterpart that propagates entire distributions rather than expectations, and that this distributional operator contracts in the Wasserstein metric.

Let $\mathcal{Z}$ denote the space of return-distribution functions mapping state-action pairs to distributions over $\mathbb{R}$. The distributional Bellman operator $\mathcal{T}^\pi$ is defined by

$$\mathcal{T}^\pi Z(s,a) \stackrel{D}{=} R(s,a) + \gamma Z(S', A'), \quad S' \sim P(\cdot|s,a),\ A' \sim \pi(\cdot|S'), \tag{1}$$

where $\stackrel{D}{=}$ denotes equality in distribution. Bellemare et al. (2017) proved that $\mathcal{T}^\pi$ is a γ-contraction in the Wasserstein distance (informally, the minimum cost of reshaping one distribution into another) between distributions,

$$\bar{d}_p(Z_1, Z_2) = \sup_{s,a} d_p\Big(Z_1(s,a),\ Z_2(s,a)\Big), \tag{2}$$

where d_p is the p-Wasserstein distance between univariate distributions. Since $\mathcal{T}^\pi$ is a contraction, iterating it converges to a unique return distribution Z^π under policy π. For quantile representations, minimizing the quantile regression loss (3) is equivalent to minimizing the 1-Wasserstein distance to the Bellman target (Dabney et al., 2018b), so QR-DQN and IQN directly inherit this convergence guarantee.[1]

A standard DQN network outputs a single scalar $Q_\theta(s,a)$ per action and minimizes the squared TD error $(r + \gamma \max_{a'} Q_{\bar{\theta}}(s', a') - Q_\theta(s,a))^2$, where r is the observed reward, s' is the next state, and $\bar{\theta}$ denotes a slowly-updated target network.

[1] $\mathcal{T}^\pi$ is not a contraction in total variation or KL divergence. The optimality operator $\mathcal{T}^*$ is not a contraction in any distribution metric, though expected values still converge (Bellemare et al., 2017).

Dabney et al. (2018b) introduced Quantile Regression DQN (QR-DQN), whose network instead outputs N values $\theta_1(s,a), \ldots, \theta_N(s,a)$ representing quantile locations of the return distribution, each with equal probability $1/N$. The quantile regression loss for a single quantile level $\tau \in (0,1)$ is $\rho_\tau(u) = u(\tau - \mathbb{1}\{u < 0\})$, which penalizes underestimates by a factor of τ and overestimates by $1-\tau$. The full QR-DQN loss sums this over all pairs of current quantiles i and target quantiles j:

$$L_{\mathrm{QR}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{N} \rho_{\hat{\tau}_i}\Big(\underbrace{r + \gamma\, \theta_j^{\mathrm{target}}(s', a^*)}_{\text{target quantile } j} - \underbrace{\theta_i(s,a)}_{\text{current quantile } i} \Big), \tag{3}$$

where $\hat{\tau}_i = (2i-1)/(2N)$ is the midpoint of the i-th quantile interval, $\theta_j^{\mathrm{target}}$ comes from the target network, and $a^* = \arg\max_{a'} \frac{1}{N} \sum_j \theta_j(s', a')$ is the greedy action under the mean return.[2]

Dabney et al. (2018a) extended this to Implicit Quantile Networks (IQN). Instead of outputting a fixed set of N quantiles, the IQN network takes a quantile level $\tau \in [0,1]$ as an additional input and outputs the corresponding return value $F_Z^{-1}(\tau; s, a)$. The IQN loss has the same pairwise structure, but with quantile levels sampled continuously:

$$L_{\mathrm{IQN}} = \frac{1}{KK'} \sum_{i=1}^{K} \sum_{j=1}^{K'} \rho_{\tau_i}\Big(r + \gamma\, F_Z^{-1}(\tau'_j; s', a^*) - F_Z^{-1}(\tau_i; s, a) \Big), \tag{4}$$

where $\tau_1, \ldots, \tau_K$ and $\tau'_1, \ldots, \tau'_{K'}$ are independent draws from $U([0,1])$, and K, K' are the number of samples per update. Because the quantile levels are sampled rather than fixed, IQN learns the full continuous quantile function rather than a discrete approximation. This continuous representation is the key bridge to risk-sensitive control.

An agent that maximizes expected return is indifferent between a certain payoff of 100 and a coin flip paying 0 or 200. In many applications this is inadequate: a portfolio manager, a robot near a cliff, or a firm facing bankruptcy all care about the shape of the return distribution, not just its mean. Distributional RL provides the return distribution; what remains is a principled way to convert that distribution into a scalar objective that encodes the desired risk attitude. Yaari (1987) showed that this can be done with a distortion function $h : [0,1] \to [0,1]$ (increasing, $h(0) = 0$, $h(1) = 1$) applied to the

[2] The resulting projected operator is a γ-contraction in the ∞-Wasserstein metric (Dabney et al., 2018b).

cumulative distribution before integrating:

$$\rho_h(Z) = \int_0^1 F_Z^{-1}(\tau)\, h'(1-\tau)\, d\tau. \tag{5}$$

This class includes the expected value ($h(\tau) = \tau$), Conditional Value-at-Risk at level α ($h(\tau) = \min(\tau/\alpha, 1)$), and the Cumulative Prospect Theory (Tversky and Kahneman, 1992), which overweights tail outcomes relative to their true probabilities. Since IQN can evaluate $F_Z^{-1}(\tau)$ at any τ, each of these risk measures reduces to choosing which quantiles to average over and how to weight them. The network itself does not change; only the sampling distribution of τ at decision time does. Sampling τ from the non-uniform distribution $\beta(\tau) = h'(1-\tau)$ rather than uniformly yields policies that maximize the distortion risk measure ρ_h.

CVaR at level α (the expected return in the worst α-fraction of outcomes) is the most widely used case. In IQN, sampling $\tau \sim U([0,\alpha])$ instead of $U([0,1])$ yields a CVaR-optimal policy. CVaR can also be optimized exactly via dynamic programming on an augmented state space (Bäuerle and Ott, 2011), or through its equivalence to robust MDPs with adversarial transition reweighting (Chow et al., 2015).[3]

0.1.1 Simulation Study: Risk-Sensitive Inventory Management

A newsvendor MDP with fat-tailed demand illustrates the mechanism. The agent manages inventory over a 10-period horizon with state $s \in \{0,\ldots,15\}$ (current stock) and action $a \in \{0,\ldots,5\}$ (order quantity). Demand follows a mixture distribution, $D \sim 0.8 \cdot \text{Poisson}(3) + 0.2 \cdot \text{Poisson}(10)$, creating occasional demand spikes that make risk preferences matter.[4] A single IQN network is trained for 50,000 episodes with $\tau \sim U([0,1])$, then evaluated under two τ-sampling distributions: $U([0,1])$ for risk-neutral and $U([0,0.05])$ for CVaR_{95}. Table 1 reports results over 50,000 evaluation episodes alongside the exact DP oracle.

The IQN-Neutral policy recovers 93% of the DP oracle's mean return. Switching to

[3] Risk-averse dynamic programming requires the risk measure to satisfy a recursive nestedness property (the multi-period risk decomposes into nested single-period evaluations, as the Bellman equation does for expected value) (Ruszczyński, 2010). Hau et al. (2023) showed that popular decompositions for CVaR are suboptimal regardless of discretization, correcting earlier claims. Tamar et al. (2015) extended the policy gradient theorem to coherent risk objectives as an alternative to the distributional approach. Prashanth et al. (2016) proved consistency of CPT-value estimation from sampled returns.

[4] Per-step reward: $5 \cdot \min(s+a, D) - 2a - 1 \cdot (s+a-D)^+ - 8 \cdot (D-s-a)^+$. The stockout penalty (8) exceeds the holding cost (1), so risk-averse agents should carry more safety stock.

Table 1: Risk-sensitive inventory management.

Policy	Mean Return	CVaR_{95}	CVaR_{99}	Avg. Order
DP Oracle	40.4	-52.6	-98.3	2.56
IQN-Neutral	37.4	-57.8	-104.0	2.44
IQN-CVaR_{95}	35.8	-55.0	-97.5	3.30

CVaR_{95} at decision time trades 1.6 points of mean return for 2.8 points of improved tail performance, achieved by ordering more safety stock (3.30 versus 2.44 units per period).

0.2 Constrained Markov Decision Processes

A constrained MDP augments the standard MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with K auxiliary cost functions $c_k : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ and constraint thresholds q_k. The agent maximizes the primary objective subject to

$$\max_{\pi} \ \mathbb{E}^{\pi}\Big[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t)\Big] \quad \text{subject to} \quad \mathbb{E}^{\pi}\Big[\sum_{t=0}^{\infty} \gamma^t c_k(S_t, A_t)\Big] \le q_k, \quad k = 1, \ldots, K. \tag{6}$$

Altman (1999) showed that this problem becomes a linear program when reformulated over *occupation measures* (the discounted fraction of time spent in each state-action pair). The occupation measure of policy π is

$$\nu^{\pi}(s, a) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t = s, A_t = a \mid \pi), \tag{7}$$

which satisfies the flow conservation constraints (probability flowing into each state from transitions equals probability flowing out via actions)

$$\sum_{a} \nu(s, a) = (1 - \gamma)\mu_0(s) + \gamma \sum_{s', a'} P(s|s', a')\nu(s', a') \quad \text{for all } s \in \mathcal{S}. \tag{8}$$

Since both the objective and constraints are linear in ν, and the feasible set forms a polytope,[5] the CMDP becomes a standard LP. By Carathéodory's theorem,[6] optimal CMDP policies mix at most $K + 1$ deterministic policies for K constraints.

[5] The feasible occupation measures form a bounded convex set (polytope) defined by non-negativity, the Bellman flow conservation equations, and the K linear constraint inequalities.

[6] Carathéodory's theorem states that any point in the convex hull of a set in $\mathbb{R}^d$ can be written as a convex combination of at most $d + 1$ extreme points. Adding K constraint inequalities introduces up to K additional dimensions along which the optimum may lie in the interior, so the optimal solution is a convex combination of at most $K + 1$ vertices.

The LP formulation yields *strong duality* under Slater's condition.[7] The dual problem is

$$\min_{\lambda\geq 0}\max_{\pi}\ L(\pi,\lambda)=\mathbb{E}^{\pi}\Big[\sum_{t=0}^{\infty}\gamma^t\Big(r(S_t,A_t)-\sum_{k=1}^{K}\lambda_k c_k(S_t,A_t)\Big)\Big]+\sum_{k=1}^{K}\lambda_k q_k, \tag{9}$$

and the optimal dual variable λ_k^* is the *shadow price* of the k-th constraint. By the envelope theorem, $\partial V^*/\partial q_k=\lambda_k^*$: the marginal change in optimal value per unit relaxation of constraint q_k. This is the same shadow price that appears in the linear programming dual of resource allocation problems.

Paternain et al. (2019) extended this result beyond the LP formulation. Despite the non-convexity of the objective in policy parameters, they proved that constrained RL has *zero duality gap*: the optimal dual value equals the primal. The proof exploits the convexity of the occupancy measure set and applies a minimax theorem. This means the CMDP can be solved exactly via dual ascent on the Lagrange multipliers, even when using parametric policy classes, with an approximation error bounded by the expressiveness of the parametrization.

Achiam et al. (2017) gave the foundational trust-region instantiation. At iteration k, the policy update solves

$$\max_{\pi}\ \mathbb{E}_{s\sim d^{\pi_k},a\sim\pi}\Big[A^{r}_{\pi_k}(s,a)\Big]\quad\text{s.t.}\quad J_{C_i}(\pi_k)+\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_k},a\sim\pi}\Big[A^{C_i}_{\pi_k}(s,a)\Big]\leq d_i,\quad \bar{D}_{\mathrm{KL}}(\pi\|\pi_k)\leq\delta, \tag{10}$$

where $A^r_{\pi_k}$ and $A^{C_i}_{\pi_k}$ are the advantage functions for the reward and the i-th cost, d^{π_k} is the discounted state visitation distribution, and $\bar{D}_{\mathrm{KL}}(\pi\|\pi_k)=\mathbb{E}_{s\sim\pi_k}[D_{\mathrm{KL}}(\pi(\cdot|s)\|\pi_k(\cdot|s))]$. For the exact solution to (10), monotone reward improvement and per-iterate constraint satisfaction are guaranteed.[8,9]

In practice, the dominant method is PPO-Lagrangian, which replaces the trust-region constraint in (10) with the PPO clipped surrogate objective applied to the Lagrangian

[7] Slater's condition requires the existence of a feasible policy π' satisfying all constraints with strict inequality: $\mathbb{E}^{\pi'}[\sum_t\gamma^t c_k(S_t,A_t)]<q_k$ for all k.

[8] Achiam et al. (2017) bound the worst-case constraint degradation at $O(\sqrt{\delta}\,\gamma\varepsilon/(1-\gamma)^2)$, where ε bounds the cost advantage, via Pinsker's inequality. The implemented Constrained Policy Optimization (CPO) algorithm solves a quadratic approximation to (10); the hard guarantee applies to the exact solution.

[9] Subsequent theoretical work sharpened convergence rates for primal-dual CMDP methods: Tessler et al. (2019) proved almost-sure convergence of a three-timescale actor-critic-multiplier scheme (RCPO) to a local Lagrangian optimum; Ding et al. (2020) established the first non-asymptotic $O(1/\sqrt{T})$ rate for both optimality gap and constraint violation (dimension-free, via Fisher information geometry); Liu et al. (2022) improved this to $O(\log(T)/T)$ using policy mirror descent; and Ying et al. (2022) achieved $O(1/T)$ with entropy regularization.

advantage $\hat{A}_t^\lambda = \hat{A}_t^r - \lambda \hat{A}_t^c$:

$$\max_\theta \ \mathbb{E}_{s,a\sim\pi_{\theta_k}}\Big[\min\Big(\rho_t \hat{A}_t^\lambda,\ \mathrm{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\,\hat{A}_t^\lambda\Big)\Big], \qquad \rho_t = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)}, \tag{11}$$

where $\hat{A}_t^r$ and $\hat{A}_t^c$ are generalized advantage estimates computed from two separate critics $V_\phi^r(s)$ and $V_\psi^c(s)$ via $\hat{A}_t = \sum_{l=0}^{T-t}(\gamma\lambda_{\mathrm{GAE}})^l \delta_{t+l}$ with TD residuals $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ (analogously for cost). After each policy update, the multiplier is adjusted by projected gradient ascent on the constraint violation:

$$\lambda_{t+1} = \Big[\lambda_t + \eta\Big(\hat{J}_C(\pi_\theta) - d\Big)\Big]_+. \tag{12}$$

Stooke et al. (2020) refined this update with a PID controller that dampens the multiplier oscillations characteristic of naive dual ascent.[10]

0.2.1 Simulation Study: Carbon-Constrained Production

A factory maximizes manufacturing profit subject to a carbon emissions budget. The state is (inventory level, demand regime), where inventory $\in \{0, \dots, 8\}$ and demand switches between low and high regimes via a Markov chain. The action is (production level, energy source), where production $\in \{0, 1, 2, 3\}$ and energy is dirty (cheap, 3.0 tons CO_2 per unit) or clean (expensive, 0.5 tons per unit). The reward is daily profit; the cost is CO_2 emitted. The carbon budget d is set to 30% of the unconstrained optimum's discounted emissions, and the LP dual yields an analytical shadow price $\lambda^* = 1.20$.[11] The CMDP is

$$\max_\pi \ \mathbb{E}^\pi\Big[\sum_{t=0}^{\infty} \gamma^t \underbrace{\Big(p\min(I_t + a_t^{\mathrm{prod}},\, D_t) - \kappa_{e_t} a_t^{\mathrm{prod}} - h\,(I_t + a_t^{\mathrm{prod}} - D_t)^+\Big)}_{r(S_t,A_t)}\Big] \quad \text{s.t.} \quad \mathbb{E}^\pi\Big[\sum_{t=0}^{\infty}\gamma^t \underbrace{\xi_{e_t}\, a_t^{\mathrm{prod}}}_{c(S_t,A_t)}\Big] \le d, \tag{13}$$

where $p = 10$ is the unit price, $\kappa_e \in \{2, 5\}$ is the production cost for dirty and clean energy respectively, $h = 1$ is the holding cost, $\xi_e \in \{3.0, 0.5\}$ is the emission rate, D_t is stochastic demand, and $\gamma = 0.95$.

Table 2 and Figure 1 compare three methods: the constrained LP oracle, uncon-

[10] The FSRL library (Liu et al., 2024) provides a pip-installable implementation of PPO-Lagrangian alongside CPO and TRPO-Lagrangian.

[11] The constrained LP over occupation measures (18 states × 8 actions = 144 variables) is solved exactly with HiGHS. The shadow price λ^* is the dual variable on the carbon constraint.

strained Q-learning, and Lagrangian Q-learning with dual ascent on (12). Each Q-learning variant is run over five seeds; the table reports means and standard errors.[12] Unconstrained Q-learning converges near the DP optimum (return 254 ± 6 versus 273) but violates the carbon budget by a factor of three (cost 95 ± 5 versus $d = 31$). The learned multiplier rises from zero and settles at $\lambda \approx 1.39$, near the analytical shadow price $\lambda^* = 1.20$. Across the five seeds the trajectory is nearly monotone: the peak value is 1.407 ± 0.003, within one percent of the settling value, so naive dual ascent here exhibits little of the overshoot that Stooke et al. (2020)'s PID controller is designed to dampen. The final λ sits about sixteen percent above λ^*, and the corresponding policy holds emissions at 23 ± 2, below the budget of 31, leaving the Lagrangian agent slightly too conservative (return 172 ± 3 versus the LP optimum of 186). The LP optimal policy is stochastic, mixing dirty and clean energy at a single state; Q-learning recovers a deterministic approximation.

Table 2: Carbon-constrained production results. Budget $d = 31.35$. Q-learning rows report means and standard errors over five seeds.

Method	Return	Cost	Budget	λ
LP Oracle	186.4	31.35	Y	1.20
Unconstrained Q-learning	253.7 ± 5.7	95.03 ± 4.65	N	–
Lagrangian Q-learning	172.5 ± 3.1	23.41 ± 2.38	Y	1.39 ± 0.00

0.3 Robust MDPs and Ambiguity Aversion

Iyengar (2005) defined the robust Bellman operator

$$(TV)(s) = \max_a \min_{p \in \mathcal{P}(s,a)} \Big\{ r(s,a) + \gamma\, p \cdot V \Big\}, \tag{14}$$

where $\mathcal{P}(s,a)$ is an uncertainty set of transition distributions for each state-action pair and $p_0(\cdot|s,a)$ is the nominal transition kernel.[13] Under the rectangularity assumption (the uncertainty sets are independent across state-action pairs, so the joint set is a Cartesian product), T is a γ-contraction in the sup-norm, so robust value iteration and policy

[12] The LP oracle value is computed by exact policy evaluation $V = (I - \gamma P_\pi)^{-1} R_\pi$, an infinite-horizon quantity. The Q-learning returns are Monte Carlo averages over rollouts truncated at horizon $H = 100$. With $\gamma^{100} \approx 6 \times 10^{-3}$, the truncation removes a non-negative discounted tail bounded by roughly $\gamma^H r_{\max}/(1-\gamma)$, on the order of one to two percent of the return. Part of the gap between the LP value and the Q-learning returns is therefore truncation bias rather than policy suboptimality.

[13] The nominal model $p_0(\cdot|s,a)$ is the agent's best estimate of the transition kernel, typically estimated from data or specified by a simulator.

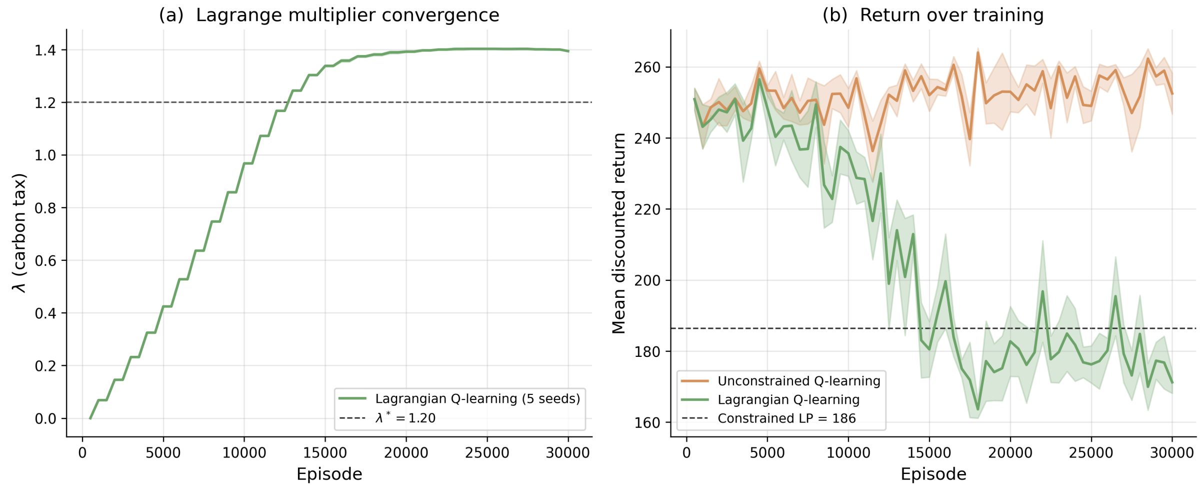

Figure 1: (a) Lagrange multiplier λ over training episodes, with the LP shadow price λ^* as the dashed reference. (b) Mean discounted return for unconstrained and Lagrangian Q-learning, with the constrained LP optimum as the dashed reference. Solid lines are means over five seeds; shaded bands are ± 1 standard error.

iteration converge with the same guarantees as their standard counterparts (recall Section **??**); the only change is substituting T for the standard Bellman operator.[14] For KL balls $\mathcal{P}(s, a) = \{p : \mathrm{KL}(p \| p_0) \leq \kappa\}$,[15] the inner minimization has the closed-form solution $q^*(s') \propto p_0(s'|s, a) \exp(-\gamma V(s')/\theta)$, an exponential tilting of nominal transitions toward low-value states, where θ is the Lagrange multiplier on the KL constraint (Nilim and El Ghaoui, 2005). Hansen and Sargent (2001) independently derived the same operator from decision theory as "multiplier preferences," in which the agent maximizes expected utility penalized by the KL cost of distorting beliefs away from a reference model. Petersen et al. (2000) proved that KL-robust MDPs, multiplier preferences, and risk-sensitive control with exponential utility $\mathbb{E}_P[\exp(-\mathrm{cost}/\theta)]$ are three descriptions of the same problem.[16]

[14] Rectangularity means $\mathcal{P} = \prod_{s,a} \mathcal{P}(s, a)$. Without it, the problem becomes NP-hard (Wiesemann et al., 2013).

[15] A KL ball of radius κ around the nominal contains all distributions "close" to p_0 in an information-theoretic sense. For example, if $p_0 = (0.1, 0.3, 0.4, 0.2)$ across four states, a ball with $\kappa = 0.1$ allows distributions like $(0.15, 0.35, 0.35, 0.15)$ but not $(0.5, 0.5, 0, 0)$, which would be too far from the nominal. Barillas et al. (2009) proposed calibrating κ (equivalently θ) via detection error probabilities. When KL sets are insufficient because p_0 assigns zero probability to relevant outcomes, Wasserstein balls $\{Q : W_p(Q, P_0) \leq \epsilon\}$ are the alternative (Mohajerin Esfahani and Kuhn, 2018; Grand-Clément and Kroer, 2021; Yu et al., 2023).

[16] See also Whittle (1981), Jacobson (1973), Fleming and McEneaney (1995). Maccheroni et al. (2006) axiomatized variational preferences, in which the agent evaluates each act under the worst-case belief penalized by a divergence cost: $V(f) = \min_p\{\int u(f)\, dp + c(p)\}$, nesting maxmin EU (Gilboa and Schmeidler, 1989), Hansen-Sargent, and mean-variance as special cases. Strzalecki (2011) showed KL is the unique penalty satisfying a natural invariance axiom; Hansen and Sargent (2024) provided a recent comprehensive treatment.

0.3.1 Algorithms for Robust RL

The robust Bellman operator (14) is a drop-in replacement for the standard Bellman target. Robust Q-learning replaces the TD target with its worst-case counterpart:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[r + \gamma \min_{p\in\mathcal{P}(s,a)} \sum_{s'} p(s') \max_{a'} Q(s',a') - Q(s,a)\Big], \tag{15}$$

where the inner minimization uses the exponential tilting for KL balls or bisection for TV and chi-squared sets.[17]

For high-dimensional problems where tabular methods and explicit uncertainty sets are infeasible, Pinto et al. (2017) introduced Robust Adversarial Reinforcement Learning (RARL):

$$\max_{\pi_{\text{agent}}} \min_{\pi_{\text{adv}}} \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t, a_t^{\text{adv}})\Big], \tag{16}$$

where a_t^{adv} is the adversary's action. Training alternates between updating the agent policy π_{agent} via TRPO or PPO with the adversary fixed, then updating the adversary π_{adv} with the agent fixed. The adversary's action space is problem-specific: joint torques for locomotion, force perturbations for manipulation. The resulting agent policies are more conservative, with wider stability margins; in locomotion tasks, robust agents adopt lower center-of-gravity gaits. Pinto et al. (2017) demonstrated that policies trained against an adversary in simulation transferred to physical robots more reliably than standard PPO policies.

Derman et al. (2021) proved that entropy regularization provides implicit robustness. The soft Bellman equation used in SAC,

$$V(s) = \tau \log \sum_a \exp\Big(Q(s,a)/\tau\Big), \tag{17}$$

is equivalent to solving a robust MDP with reward uncertainty set $\{r' : \mathrm{KL}(r'\|r) \le \kappa\}$, where the entropy temperature τ maps directly to the robustness radius κ. Adding a value-regularization term extends this to robustness against transition misspecification.[18]

[17] Sample complexity for learning robust policies scales polynomially in $|\mathcal{S}||\mathcal{A}|$: Panaganti and Kalathil (2022) established the first bounds under (s,a)-rectangular uncertainty for TV, KL, and chi-squared sets; Clavier et al. (2024) improved the rates for model-based approaches.

[18] The "Twice Regularized MDP" (R^2-MDP) combines policy regularization (reward robustness) with value regularization (transition robustness). Standard SAC training already implements the reward-robust component; transition robustness requires an additional penalty on the divergence between learned and nominal dynamics

These three approaches offer different trade-offs. Robust Q-learning requires an explicit uncertainty set but provides exact minimax guarantees for tabular problems. RARL avoids specifying an uncertainty set by learning the adversary, making it practical for continuous control, but provides no formal robustness certificate. Entropy regularization provides implicit robustness at zero additional implementation cost for practitioners already using SAC, but ties the robustness radius to the temperature parameter.

0.3.2 Simulation Study: Consumption-Savings Under Model Mismatch

A consumption-savings agent with constant relative risk aversion (CRRA) utility $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma = 2$, receives stochastic income $y \in \{1, \dots, 5\}$ each period and chooses how much to consume versus save at gross return $R = 1.02$, with $\gamma = 0.95$. The robust Bellman equation for this problem is

$$V(w) = \max_{c \in \{0,\dots,w\}} \Big\{ u(c) + \gamma \min_{q:\, \mathrm{KL}(q \| p_0) \le \kappa} \sum_y q(y)\, V\Big(R(w-c)+y\Big) \Big\}, \tag{18}$$

where w is wealth, c is consumption, and the inner minimization tilts income probabilities toward low-income states via $q^*(y) \propto p_0(y) \exp(-\gamma V(R(w-c)+y)/\theta)$. Seven policies are computed: standard DP under the nominal income distribution $p_0 = (0.05, 0.10, 0.20, 0.30, 0.35)$, robust DP at $\theta = 5$ (moderate) and $\theta = 2$ (high robustness), an oracle that knows the true perturbed distribution $\tilde{p} = (0.30, 0.30, 0.20, 0.10, 0.10)$, and three model-free counterparts trained via tabular Q-learning under the nominal model.[19] All seven policies are then evaluated under both the nominal and perturbed income models.

Table 3 reports discounted returns under both models. Under the nominal model, all policies perform similarly. Under the perturbed model, standard DP degrades by 76% while the $\theta = 2$ robust policy degrades by 72%. The Q-learning and robust Q-learning policies match their DP counterparts, confirming that the model-free algorithms recover the same precautionary savings behavior. Figure 2 shows the mechanism: robust agents consume less at each wealth level, building a precautionary savings buffer against adverse income realizations.

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models.

[19] Standard Q-learning uses the usual TD target $r + \gamma \max_{a'} Q(s', a')$; robust Q-learning replaces this with the worst-case target (15), using the nominal kernel for the inner minimization (generative model setting). Visit-count learning rates $\alpha = C/(C + N(s,a))$ with $C = 100$ and ε-greedy exploration decaying from 1.0 to 0.05 over 10^5 episodes.

Table 3: Discounted returns under nominal and perturbed income. DP policies are computed via value iteration; Q-learning policies are trained under the nominal model.

Method	Nominal	Perturbed	Degradation (%)
Standard DP	-4.94	-8.69	-76.0
Q-learning	-4.94	-8.70	-76.0
Robust DP (θ=5.0)	-4.94	-8.67	-75.5
Robust Q-learning (θ=5)	-4.95	-8.68	-75.5
Robust DP (θ=2.0)	-4.95	-8.49	-71.6
Robust Q-learning (θ=2)	-4.95	-8.49	-71.5
Oracle	-5.46	-7.60	-39.1

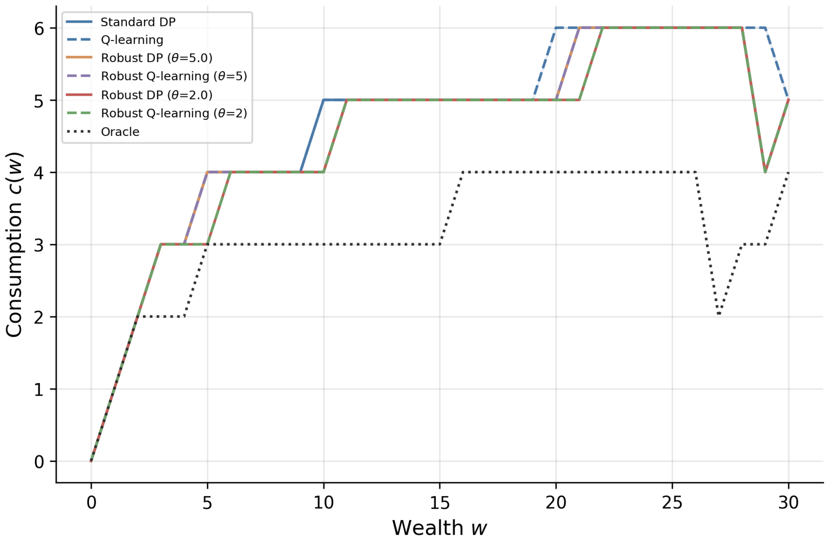

Figure 2: Consumption policy $c(w)$ for each method. Dashed lines show Q-learning policies; solid lines show DP.

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